

TLA+ Study Group — Week 04

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Brief Description of the Topics

- Description of PAXOS a fault-tolerant distributed transaction-commit algorithm

New syntax:

CHOOSE `CHOOSE v \in S : P equals`

if S is a singleton set then v is the value of the element of the singleton set; otherwise v is unspecified but P holds. If P cannot hold TLA will mark an error. One should only use it in the case of singletons or when, in a larger expression, the value doesn't depend on which v was chosen

LET `LET value == $expr`
 `IN $expr`

set comprehension $v \in S : P$ `{`
 `{ v \in S : P }`

subset of S consisting of all v satisfying P for example:

`{ n \in Nat : n > 17 } (* = { 18, 19, 20 ... }`

set comprehension $e : v \in S$ `{`
 `{ e : v \in S }`

for example

`{n^2 : n \in Nat} (* { 0, 1, 4, 9... }`

generalization of EXECPT `aState' = [aState EXCEPT ![m.ins][acc].mbal = m.bal,`
 `![m.ins][acc].bal = m.bal,`
 `![m.ins][acc].val = m.val`

is equivalent to

```
aState[m.ins][acc].mbal = m.bal;  
aState[m.ins][acc].bal = m.bal;  
aState[m.ins][acc].val = m.val;
```

i.e a "copy method" for records that allows for arbitrary nesting and chaining

What stood out

Open Questions